

Diet and gallbladder cancer: a case-control study.



Cancer of the gallbladder is rare but fatal, and has an unusual geographic and demographic distribution. Gallstones and obesity have been suggested as possible risk factors. As diet is known to influence both these factors, we carried out the present study to evaluate the possible role of diet in gallbladder carcinogenesis. A case-control study involving 64 newly diagnosed cases of gallbladder cancer and 101 cases of gallstones was carried out. The dietary evaluation was carried out by the dietary recall method based on a preset questionnaire developed specifically for the present study, keeping in mind the common dietary habits prevailing in this part of the world. Odds ratios (OR) and 95% confidence interval (CI) were calculated for various dietary items. A significant reduction in odds ratio was seen with the consumption of radish (OR 0.4; 95% CI 0.17-0.94), green chilli (OR 0.45; 95% CI 0.21-0.94) and sweet potato (OR 0.33; 95% CI 0.13-0.83) among vegetables, and mango (OR 0.4; 95% CI 0.16-0.99), orange (OR; 0.45; 95% CI 0.22-0.93), melon (OR 0.3; 95% CI 0.14-0.64) and papaya (OR 0.44; 95% 0.2-0.64) among fruits. A reduction in odds was also seen with the consumption of cruciferous vegetables, beans, onion and turnip, however the difference was not statistically significant. On the other hand, an increase in the odds was observed with consumption of capsicum (OR 2.2), beef (OR 2.58), tea (OR 1.98), red chilli (OR 1.29) and mutton (OR 1.2), however the difference was statistically not significant. In conclusion, the results of the present study show a protective effect of vegetables and fruits on gallbladder carcinogenesis, but red meat (beef and mutton) was found to be associated with increased risk of gallbladder cancer.